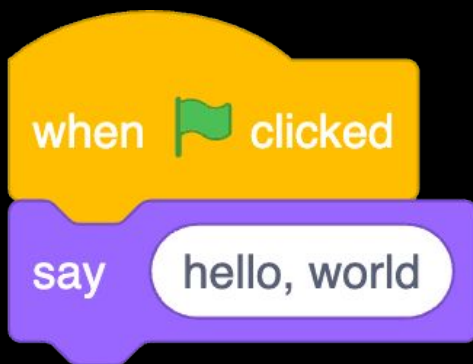


**This is CS50**

GETTING INFORMATION  
FROM **NOT**  
OLIVE TRYING TO  
GET A DRINK  
FROM A  
FIRE HOSE...





```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    printf("hello, world\n");
```

```
}
```

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    printf("hello, world\n");
```

```
}
```

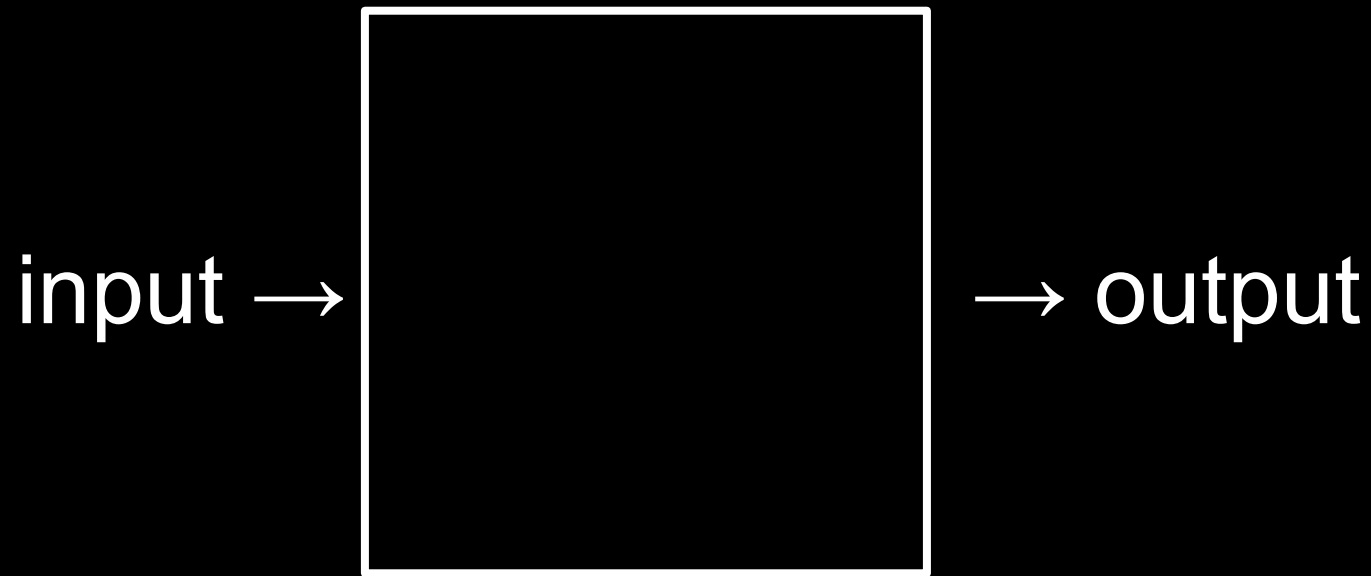
source code

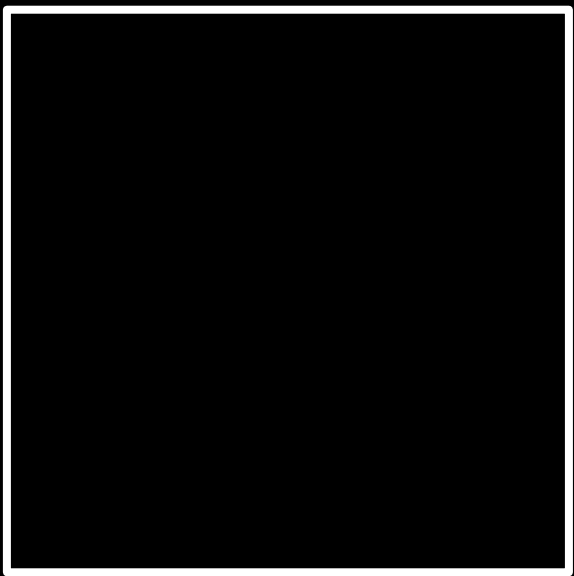
|          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|
| 01111111 | 01000101 | 01001100 | 01000110 | 00000010 | 00000001 | 00000001 | 00000000 |
| 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 00000010 | 00000000 | 00111110 | 00000000 | 00000001 | 00000000 | 00000000 | 00000000 |
| 10110000 | 00000101 | 01000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 01000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 11010000 | 00010011 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 00000000 | 00000000 | 00000000 | 00000000 | 01000000 | 00000000 | 00111000 | 00000000 |
| 00001001 | 00000000 | 01000000 | 00000000 | 00100100 | 00000000 | 00100001 | 00000000 |
| 00000110 | 00000000 | 00000000 | 00000000 | 00000101 | 00000000 | 00000000 | 00000000 |
| 01000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 01000000 | 00000000 | 01000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 01000000 | 00000000 | 01000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 11111000 | 00000001 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 11111000 | 00000001 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 00001000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |
| 00000011 | 00000000 | 00000000 | 00000000 | 00000100 | 00000000 | 00000000 | 00000000 |
| 00111000 | 00000010 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 | 00000000 |

...

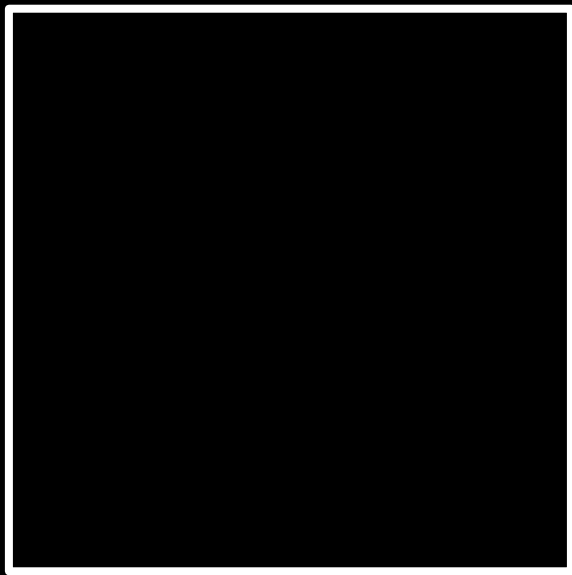
machine code



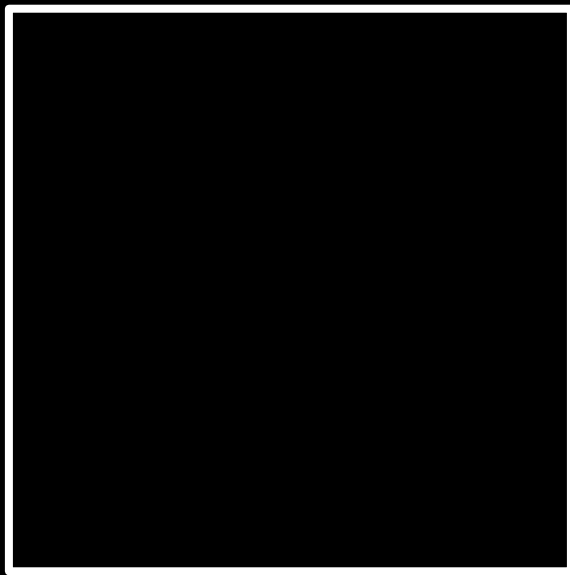




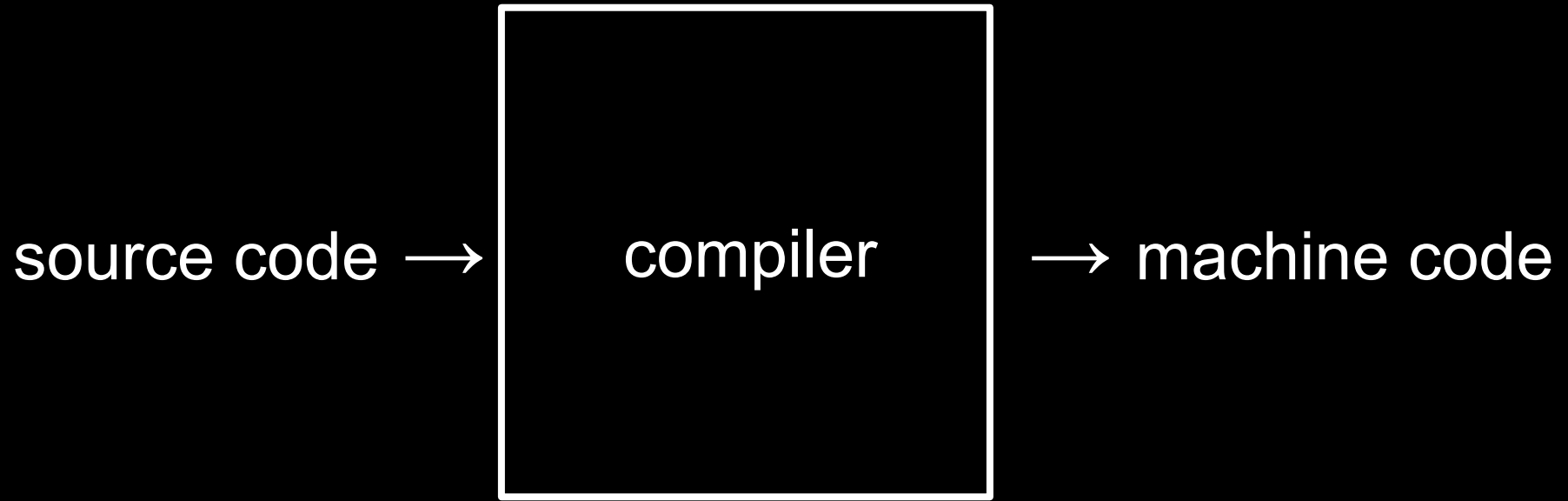
source code →



source code →



→ machine code



# Visual Studio Code for CS50

cs50.dev



EXPLORER



hello.c



✓ HELLO [CODESPACES]

hello.c



&gt; OUTLINE

&gt; TIMELINE

```
1 #include <stdio.h>
2
3 int main(void)
4 {
5     printf("hello, world\n");
6 }
```

TERMINAL



\$ make hello



EXPLORER



✓ HELLO [CODESPACES]

hello.c



&gt; OUTLINE

&gt; TIMELINE



hello.c



```
1 #include <stdio.h>
2
3 int main(void)
4 {
5     printf("hello, world\n");
6 }
```

TERMINAL



\$ make hello





EXPLORER



✓ HELLO [CODESPACES]

hello.c



&gt; OUTLINE

&gt; TIMELINE



hello.c



```
1 #include <stdio.h>
2
3 int main(void)
4 {
5     printf("hello, world\n");
6 }
```

TERMINAL



\$ make hello



EXPLORER



✓ HELLO [CODESPACES]

hello.c



&gt; OUTLINE

&gt; TIMELINE



hello.c



```
1 #include <stdio.h>
2
3 int main(void)
4 {
5     printf("hello, world\n");
6 }
```

TERMINAL



\$ make hello



EXPLORER



hello.c



✓ HELLO [CODESPACES]

hello.c



&gt; OUTLINE

&gt; TIMELINE

```
1 #include <stdio.h>
2
3 int main(void)
4 {
5     printf("hello, world\n");
6 }
```

TERMINAL



\$ make hello



EXPLORER



hello.c



✓ HELLO [CODESPACES]

hello.c



&gt; OUTLINE

&gt; TIMELINE

```
1 #include <stdio.h>
2
3 int main(void)
4 {
5     printf("hello, world\n");
6 }
```

TERMINAL



\$ make hello

graphical user interface

GUI



EXPLORER



✓ HELLO [CODESPACES]

hello.c



&gt; OUTLINE

&gt; TIMELINE



hello.c



```
1 #include <stdio.h>
2
3 int main(void)
4 {
5     printf("hello, world\n");
6 }
```

TERMINAL



\$ make hello

command-line interface



CLI



EXPLORER



hello.c



✓ HELLO [CODESPACES]

hello.c



&gt; OUTLINE

&gt; TIMELINE

```
1 #include <stdio.h>
2
3 int main(void)
4 {
5     printf("hello, world\n");
6 }
```

TERMINAL



\$ make hello

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

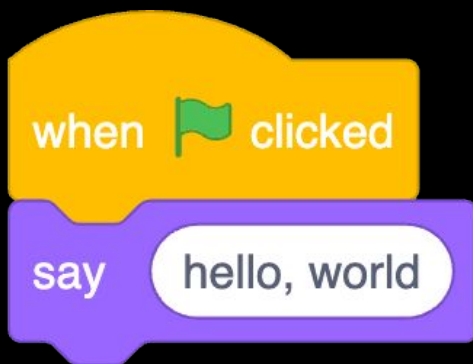
```
    printf("hello, world\n");
```

```
}
```

```
code hello.c
```

```
make hello
```

```
./hello
```



```
#include <stdio.h>
```

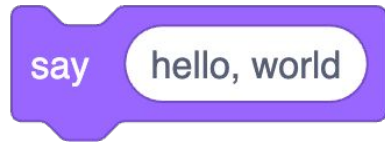
```
int main(void)
```

```
{
```

```
    printf("hello, world\n");
```

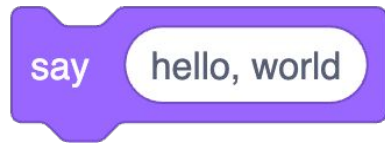
```
}
```



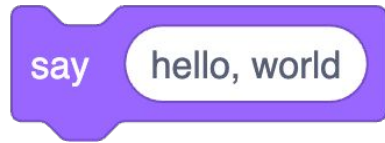


```
print (
```

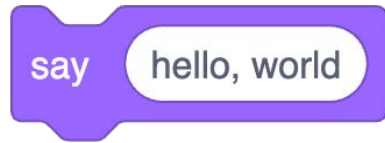




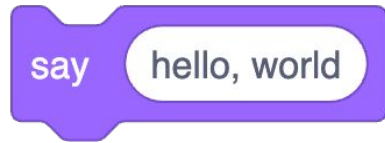
```
printf(      )
```



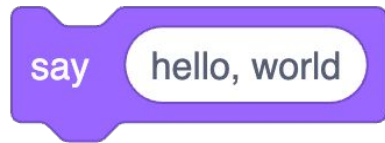
```
printf( hello, world  )
```



```
printf("hello, world ")
```



```
printf("hello, world\n")
```

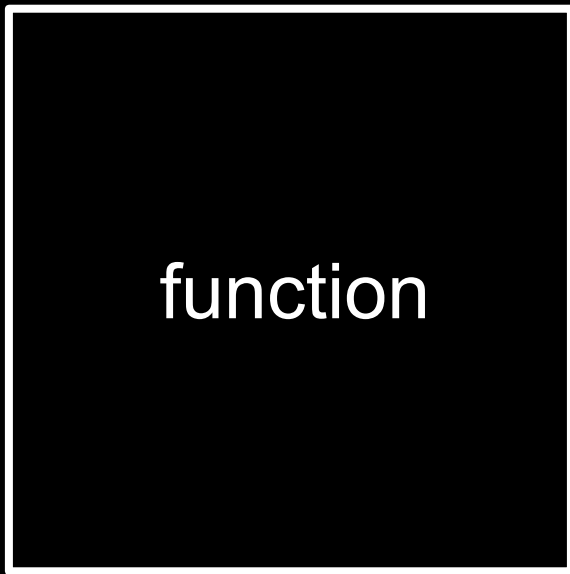


```
printf("hello, world\n");
```

$$f(x)$$



arguments →

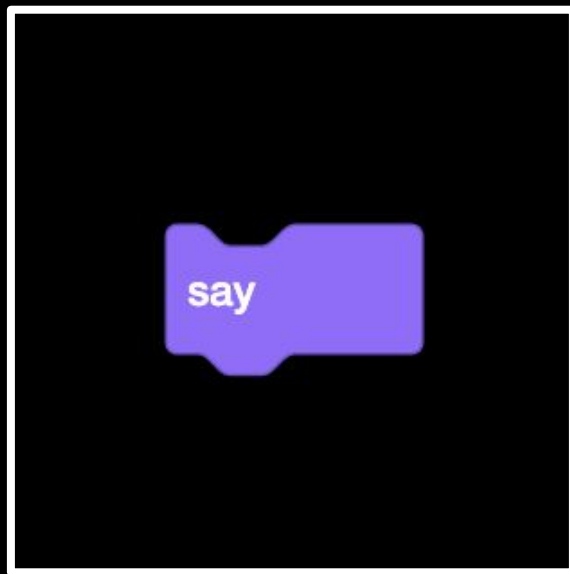


function

→ side effects



hello, world



escape sequences

`\n`

`\r`

`\"`

`\'`

`\\`

`...`

header files

libraries

stdio.h

# Manual Pages

[manual.cs50.io](http://manual.cs50.io)

[manual.cs50.io/#stdio.h](http://manual.cs50.io/#stdio.h)



[manual.cs50.io/3/printf](http://manual.cs50.io/3/printf)

cs50.h

[manual.cs50.io/#cs50.h](http://manual.cs50.io/#cs50.h)

get\_char

get\_double

get\_float

get\_int

get\_long

get\_string

...

get\_char

get\_double

get\_float

get\_int

get\_long

get\_string

...

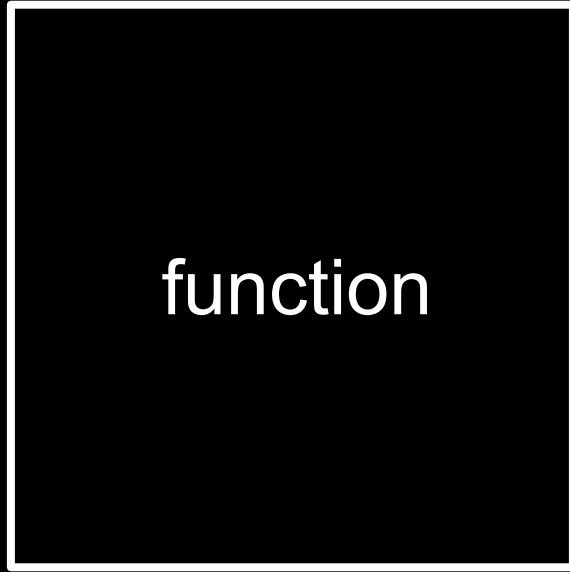
when  clicked

ask What's your name? and wait

say join hello, answer



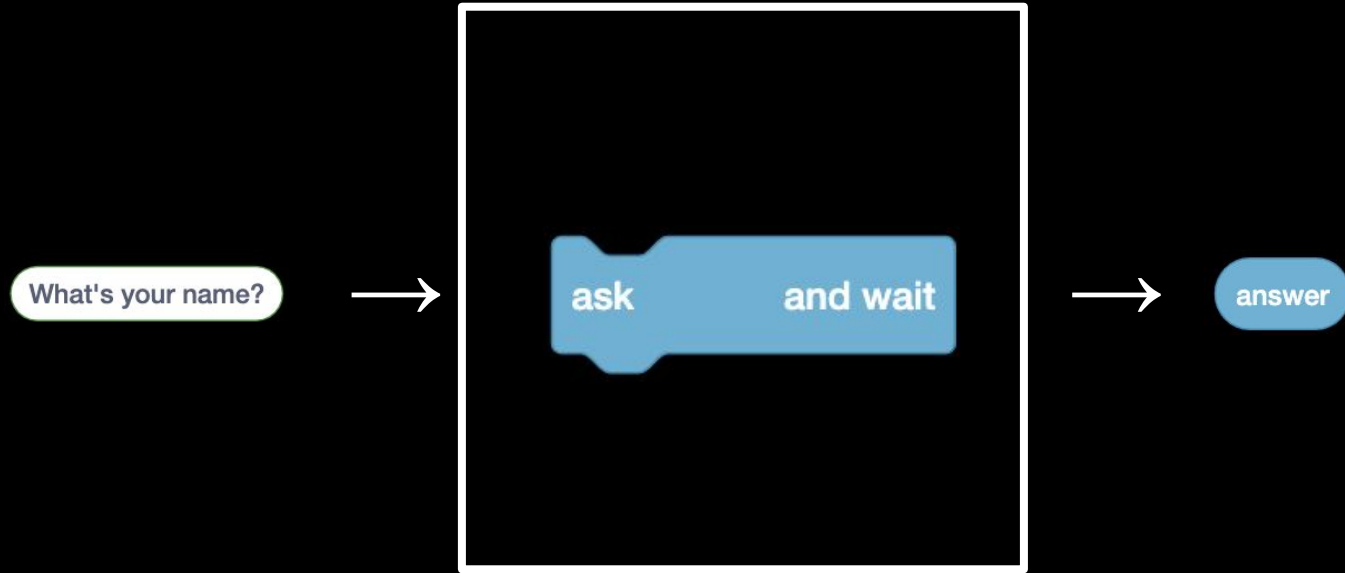
arguments →



function

→ return value





ask

What's your name?

and wait

answer

ask

What's your name? and wait

answer

```
get_string( )
```

ask What's your name? and wait

answer

```
get_string( What's your name? )
```

ask What's your name? and wait

answer

```
get_string("What's your name? ")
```

ask What's your name? and wait

answer

```
answer = get_string("What's your name? ")
```

ask What's your name? and wait

answer

```
string answer = get_string("What's your name? ")
```

ask What's your name? and wait

answer

```
string answer = get_string("What's your name? ");
```







```
printf( );
```



```
printf( hello, %s      );
```



```
printf("hello, %s  "      );
```



```
printf("hello, %s\n"      );
```



```
printf("hello, %s\n", answer);
```

Linux

cd

cp

ls

mkdir

mv

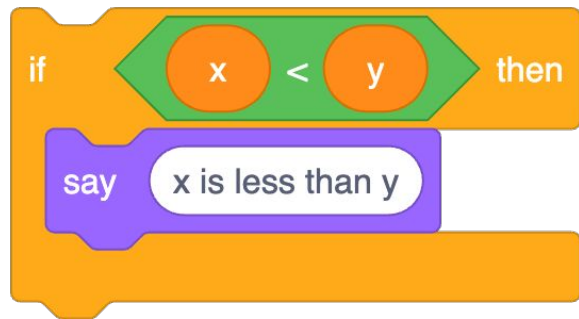
rm

rmdir

...



conditionals

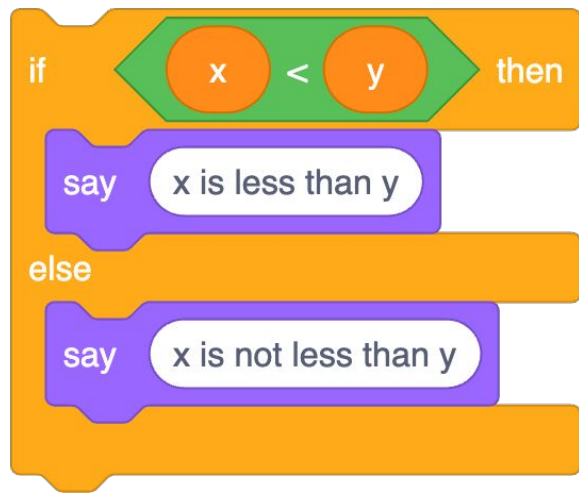


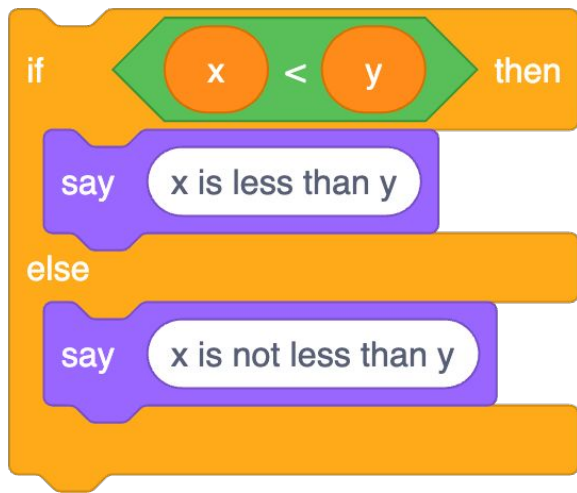


```
if (x < y)
{
}
}
```

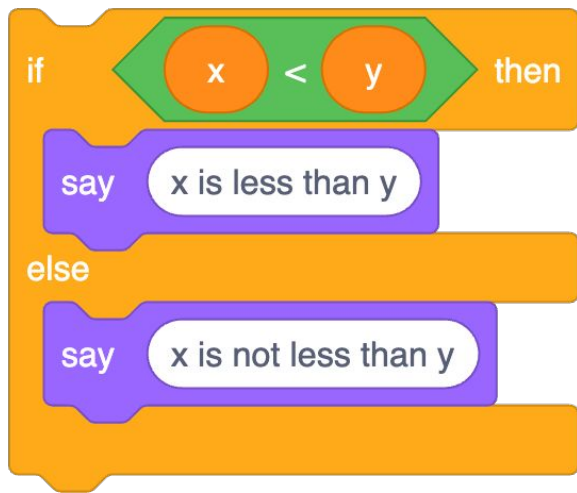


```
if (x < y)
{
    printf("x is less than y\n");
}
```

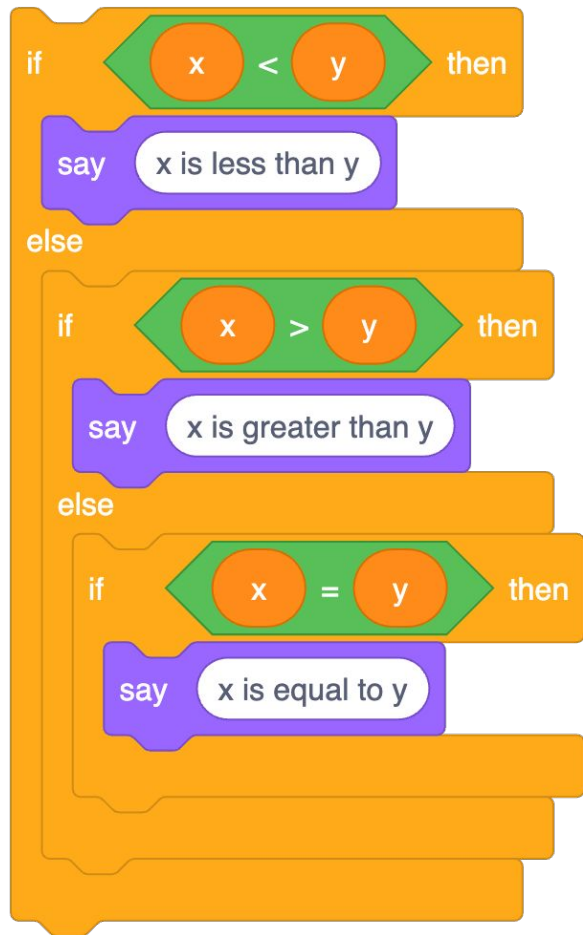




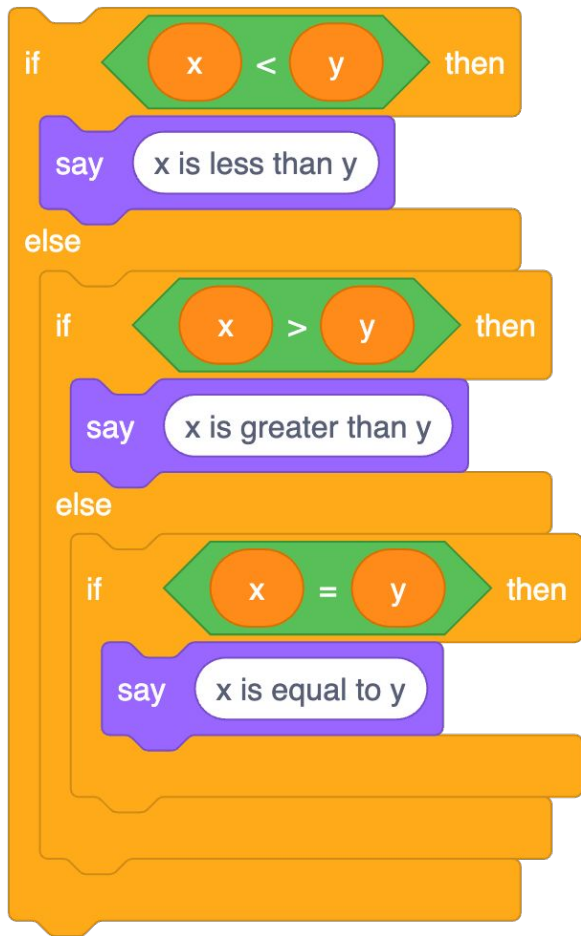
```
if (x < y)
{
}
else
{
}
```



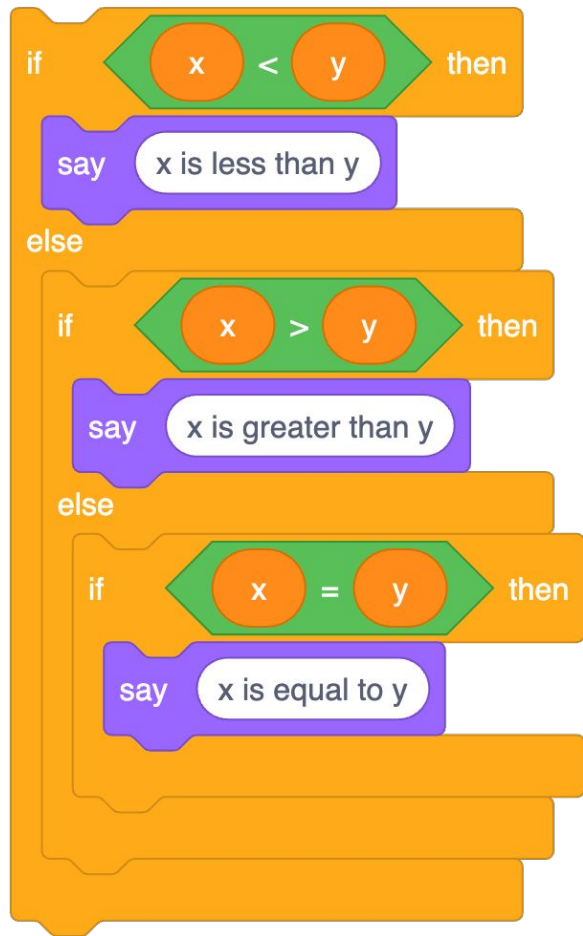
```
if (x < y)
{
    printf("x is less than y\n");
}
else
{
    printf("x is not less than y\n");
}
```



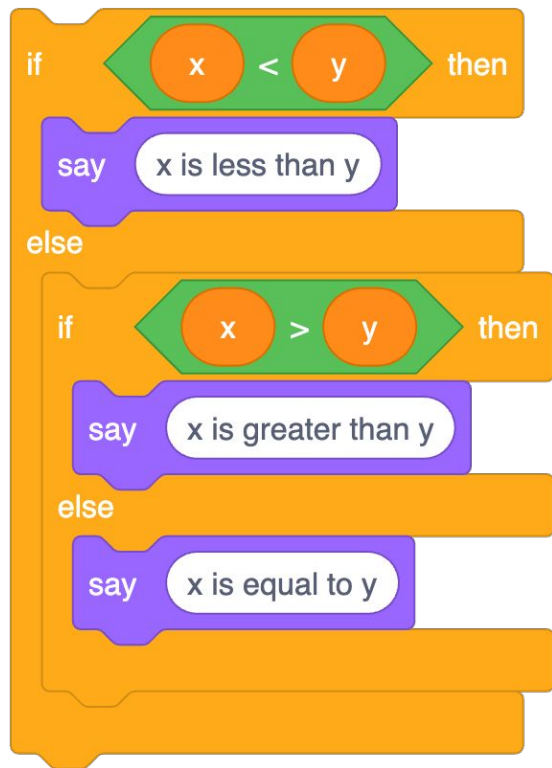




```
if (x < y)
{
}
else if (x > y)
{
}
else if (x == y)
{
}
```



```
if (x < y)
{
    printf("x is less than y\n");
}
else if (x > y)
{
    printf("x is greater than y\n");
}
else if (x == y)
{
    printf("x is equal to y\n");
}
```



```
if (x < y)
{
    printf("x is less than y\n");
}
else if (x > y)
{
    printf("x is greater than y\n");
}
else
{
    printf("x is equal to y\n");
}
```

operators

=

<

<=

>

>=

==

!=

...

types

bool

char

double

float

int

long

string

...

bool

char

double

float

int

long

string

...



cs50.h

get\_char

get\_double

get\_float

get\_int

get\_long

get\_string

...

get\_char

get\_double

get\_float

get\_int

get\_long

get\_string

...

format codes

%c

%f

%i

%li

%s

%c

%f

%i

%li

%s

variables







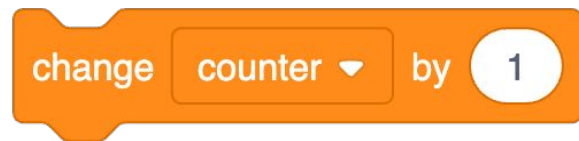
```
counter = 0
```

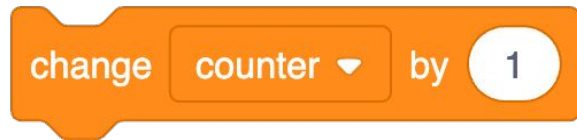


```
int counter = 0
```

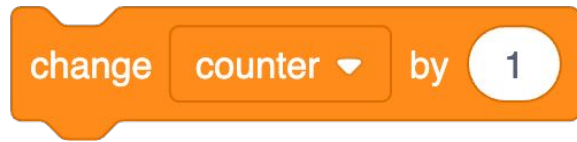


```
int counter = 0;
```

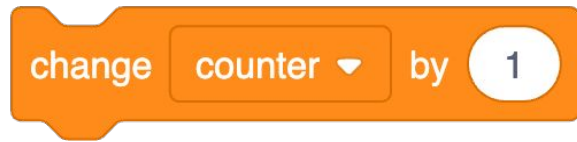




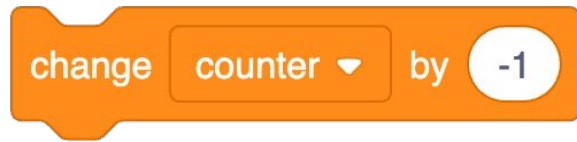
```
counter = counter + 1;
```



```
counter += 1;
```

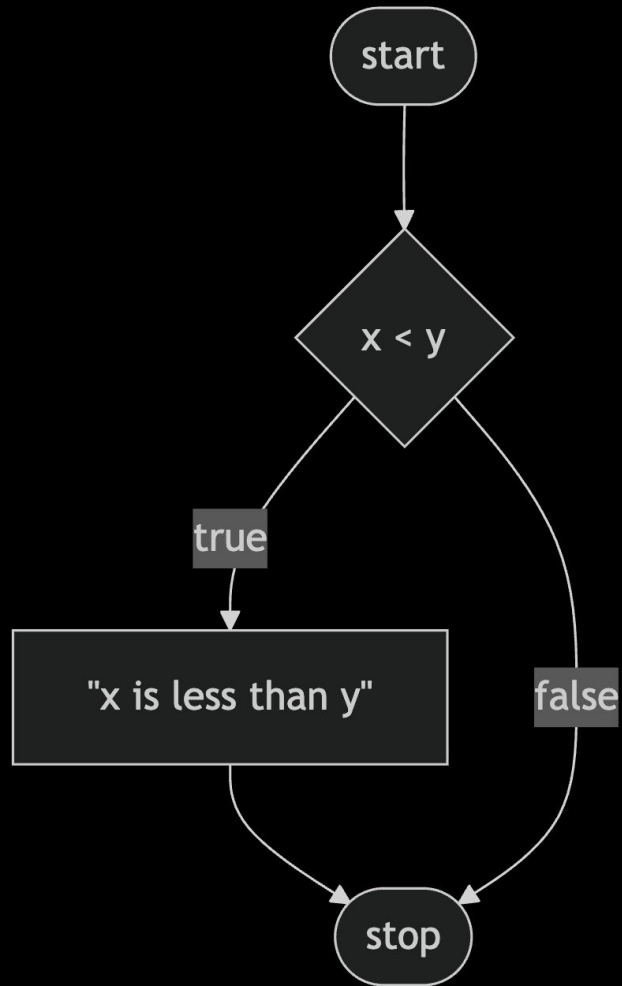


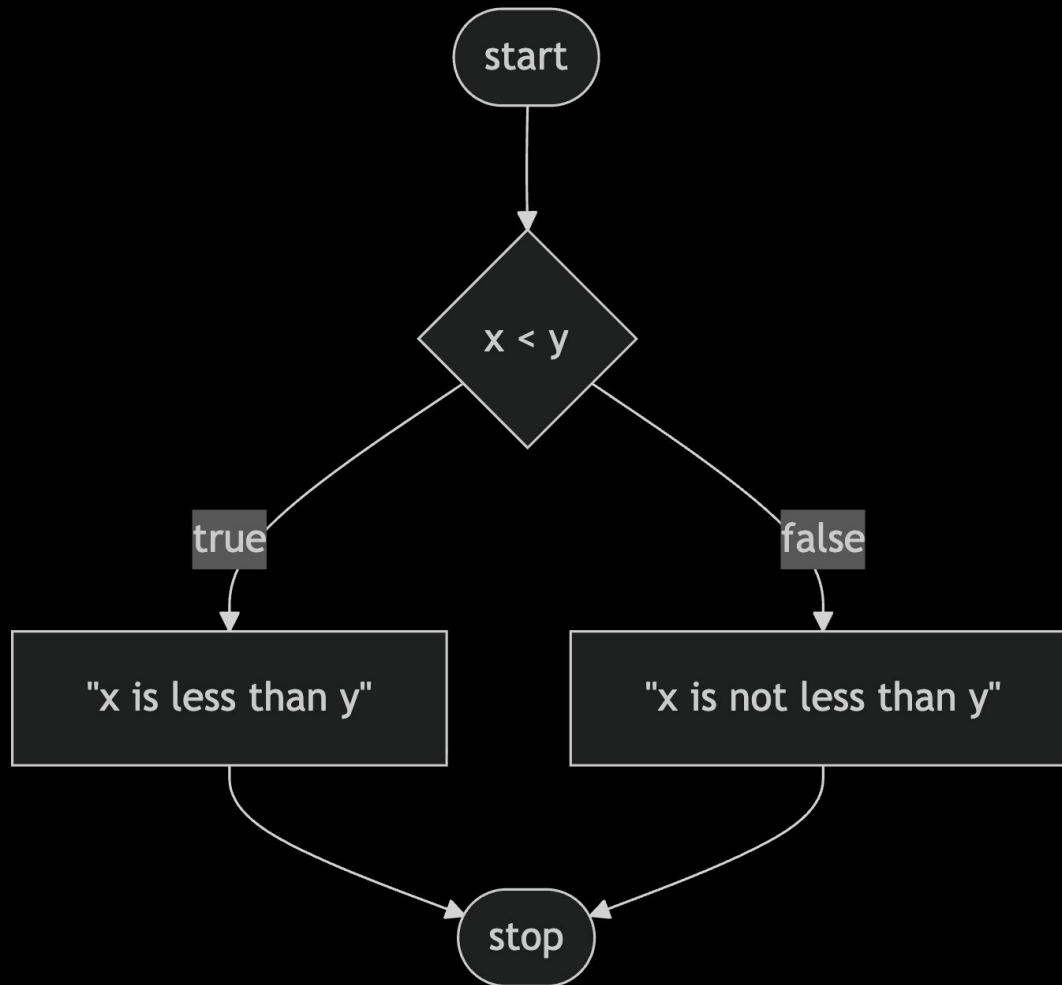
```
counter++;
```

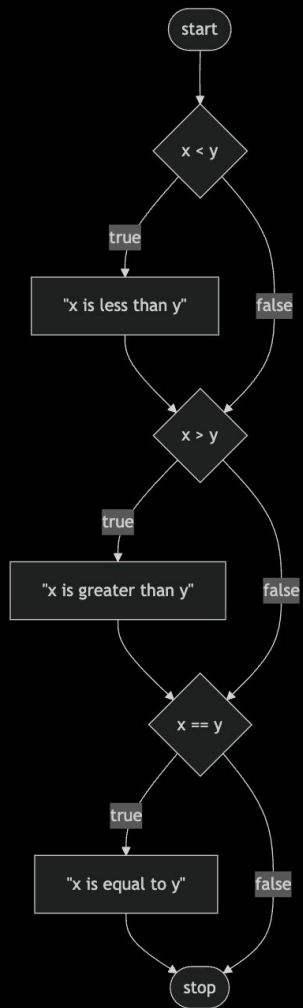


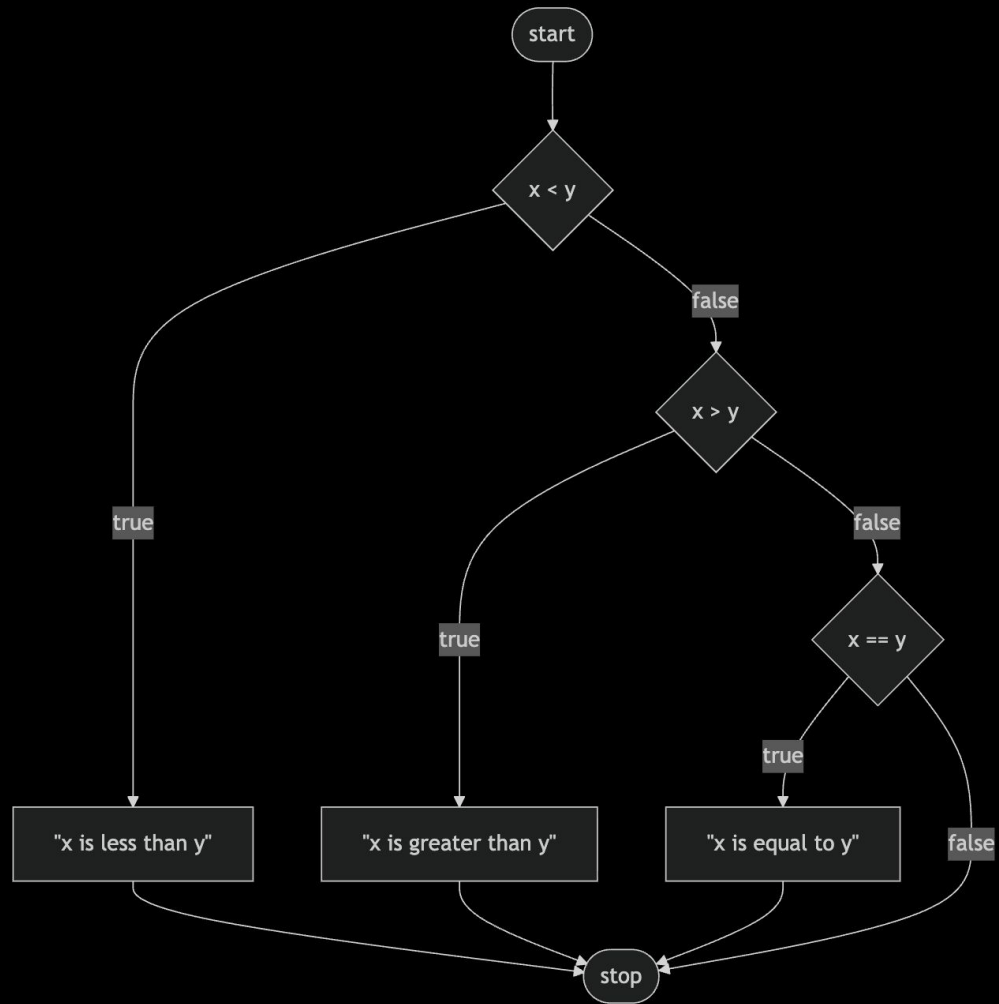
```
counter--;
```

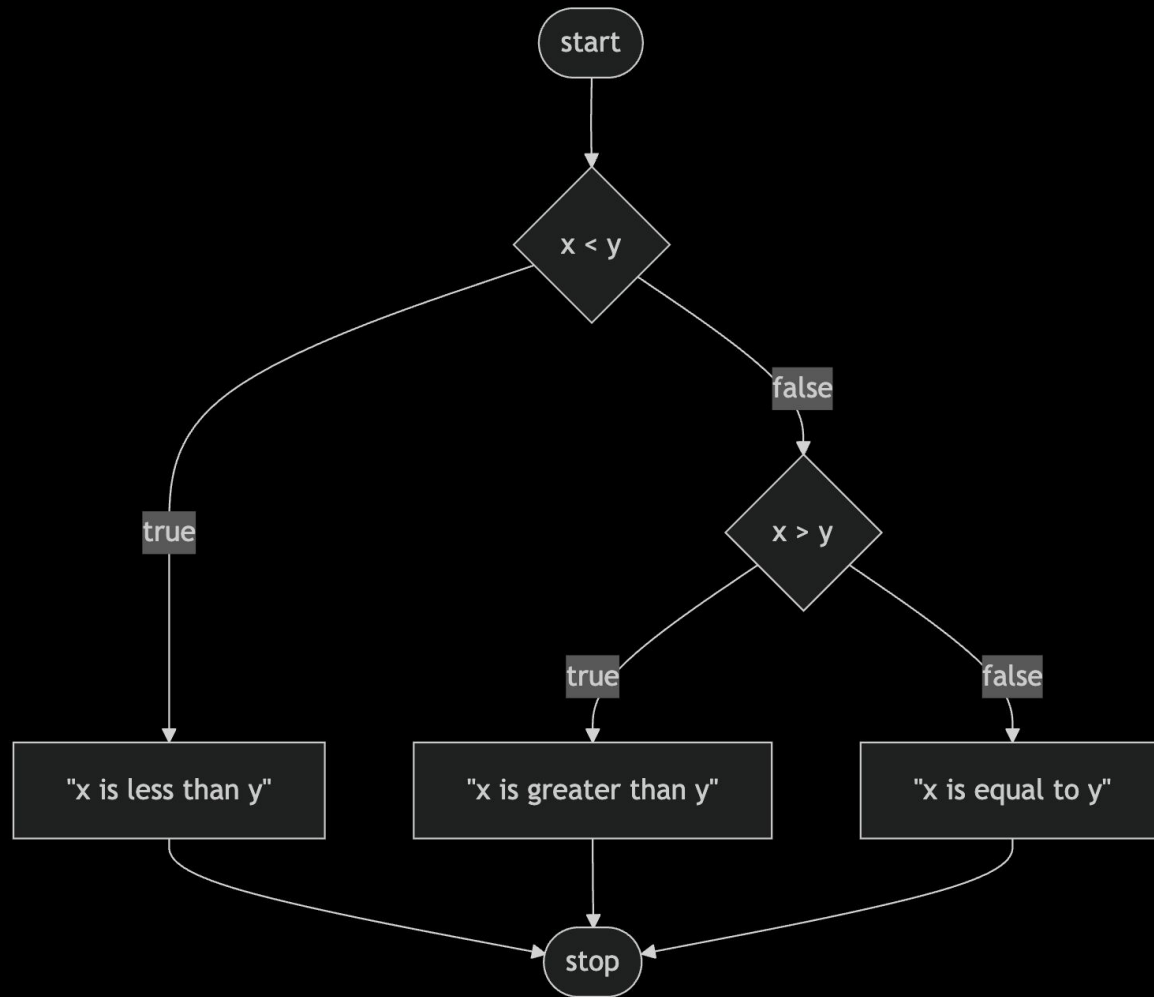












bool

char

double

float

int

long

string

...

get\_char

get\_double

get\_float

get\_int

get\_long

get\_string

...

loops







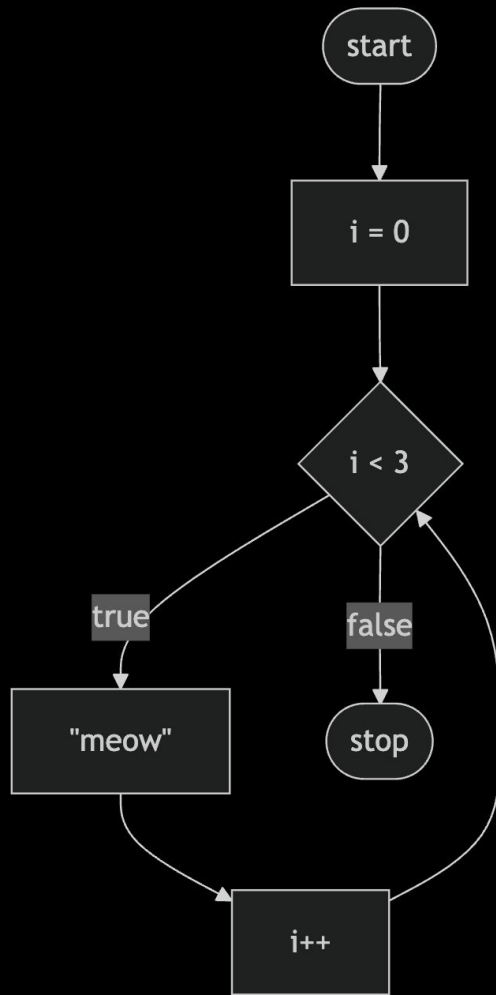
```
int i = 3;
while (i > 0)
{
    i--;
}
```



```
int i = 3;  
while (i > 0)  
{  
    printf("meow\n");  
    i--;  
}
```

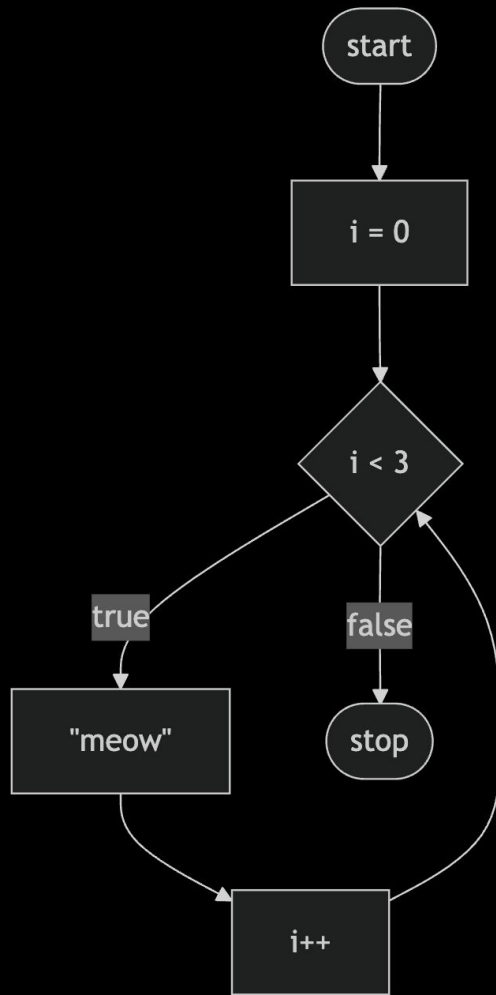


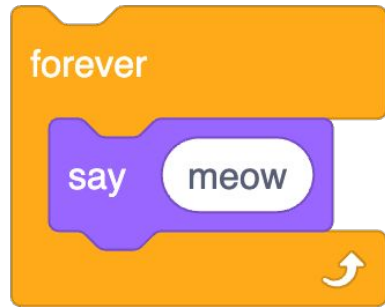
```
int i = 0;  
while (i < 3)  
{  
    printf("meow\n");  
    i++;  
}
```



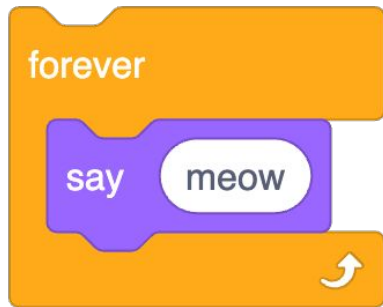


```
for (int i = 0; i < 3; i++)  
{  
    printf("meow\n");  
}
```

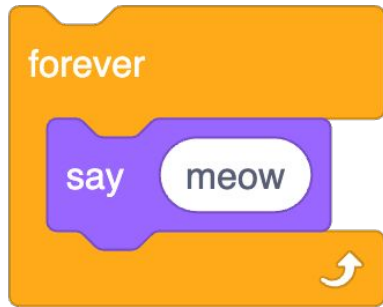




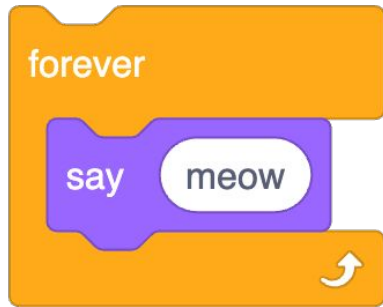




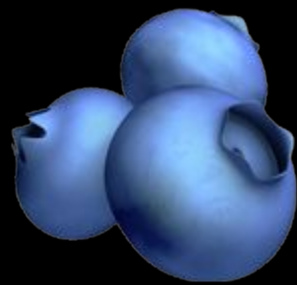
```
while (    )  
{  
  
}
```



```
while (true)
{
}
```



```
while (true)
{
    printf("meow\n");
}
```



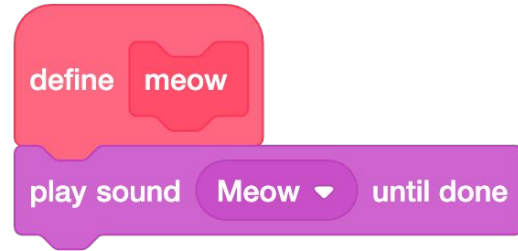
control-c

scope

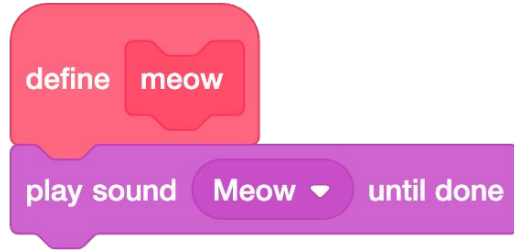
```
while (not edge) {  
  run();  
}
```

```
do {  
  run();  
} while (not edge);
```







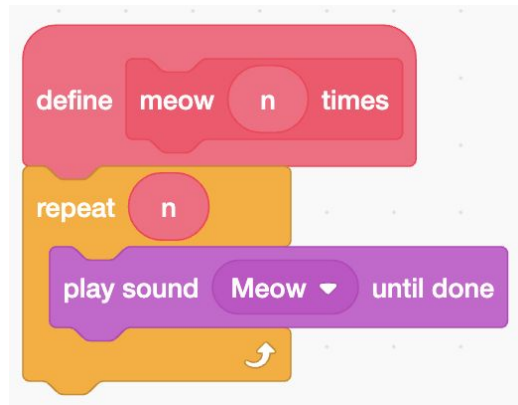


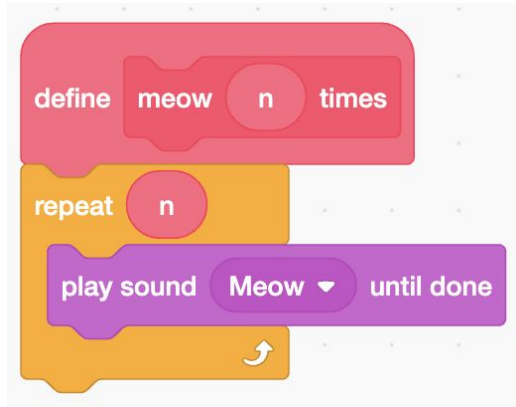
```
void meow(void)
{
    printf("meow\n");
}
```





```
int main(void)
{
    for (int i = 0; i < 3; i++)
    {
        meow();
    }
}
```





```
void meow(int n)
{
    for (int i = 0; i < n; i++)
    {
        printf("meow\n");
    }
}
```





```
int main(void)
{
    meow(3);
}
```

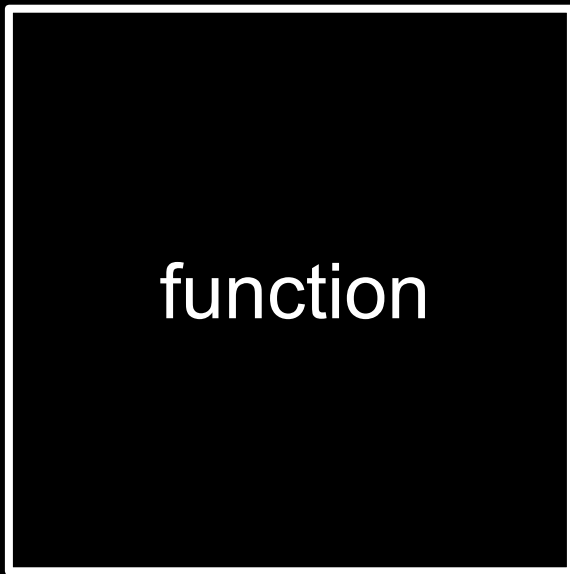
scope



correctness, design, style

check50, design50, style50

arguments →



function

→ side effects

MARIO  
000000

● × 00

WORLD  
1-1

TIME

# SUPER MARIO BROS.

©1985 NINTENDO

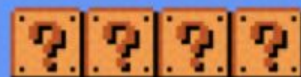


1 PLAYER GAME

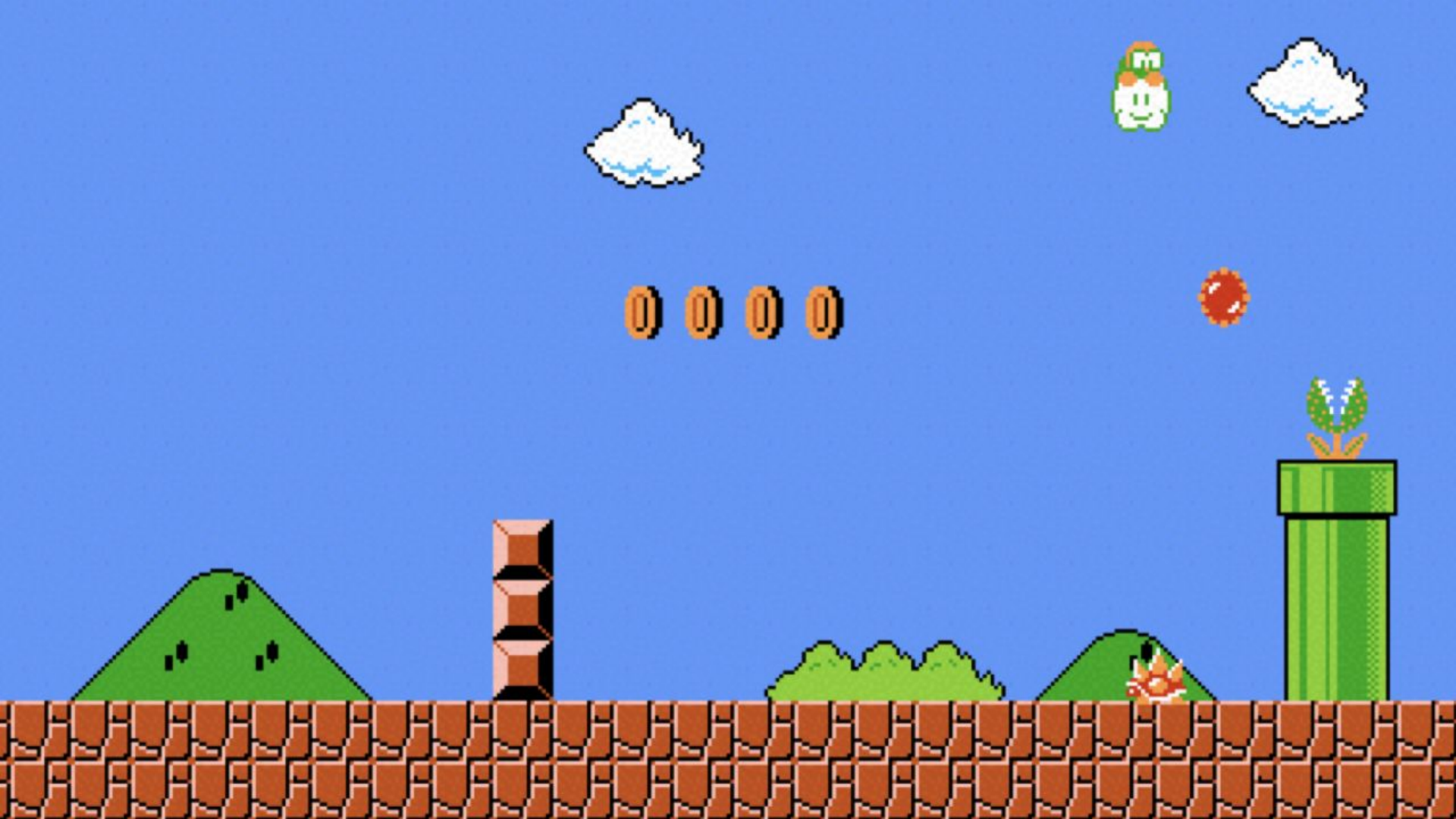
2 PLAYER GAME

TOP- 000000

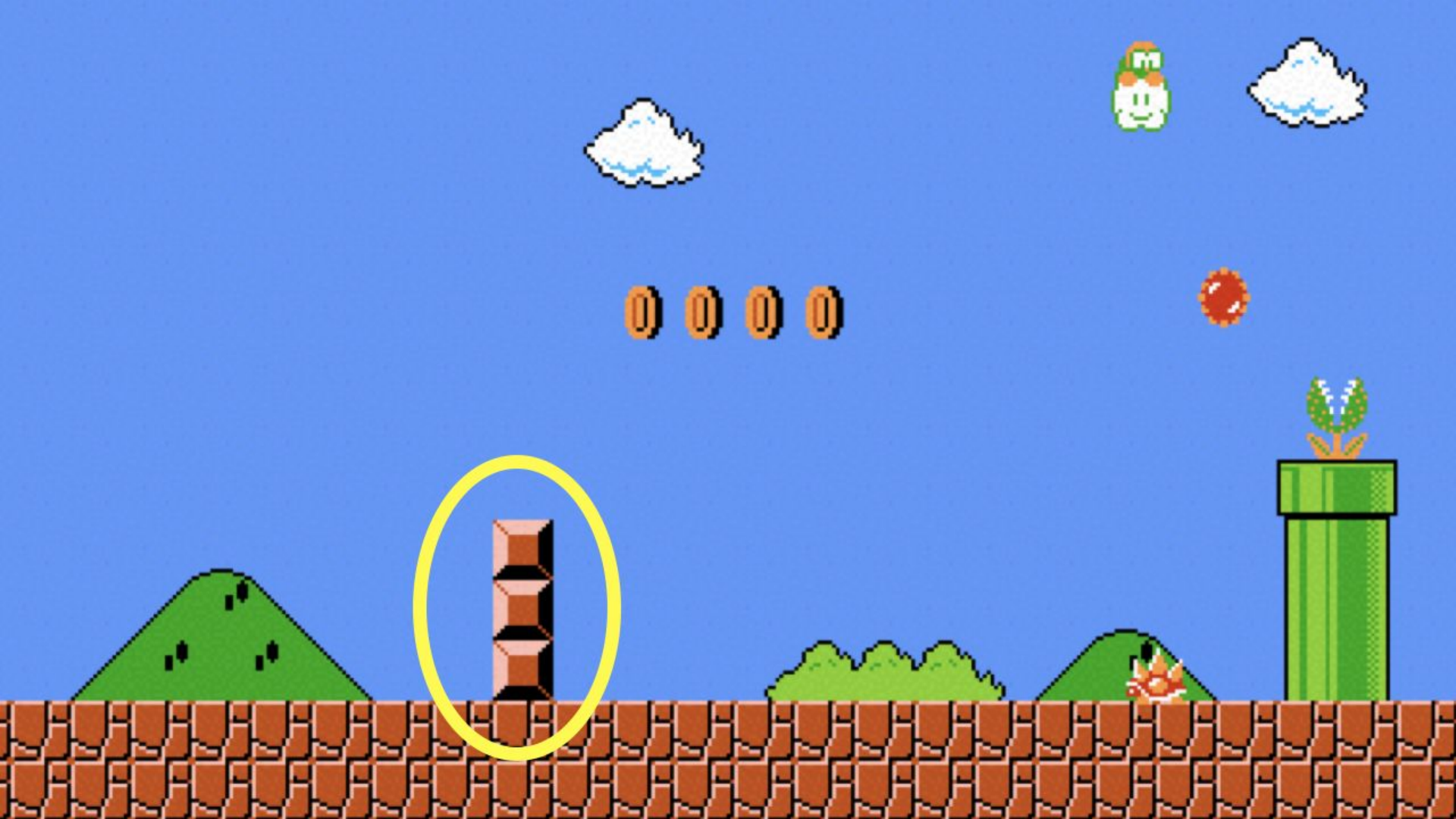




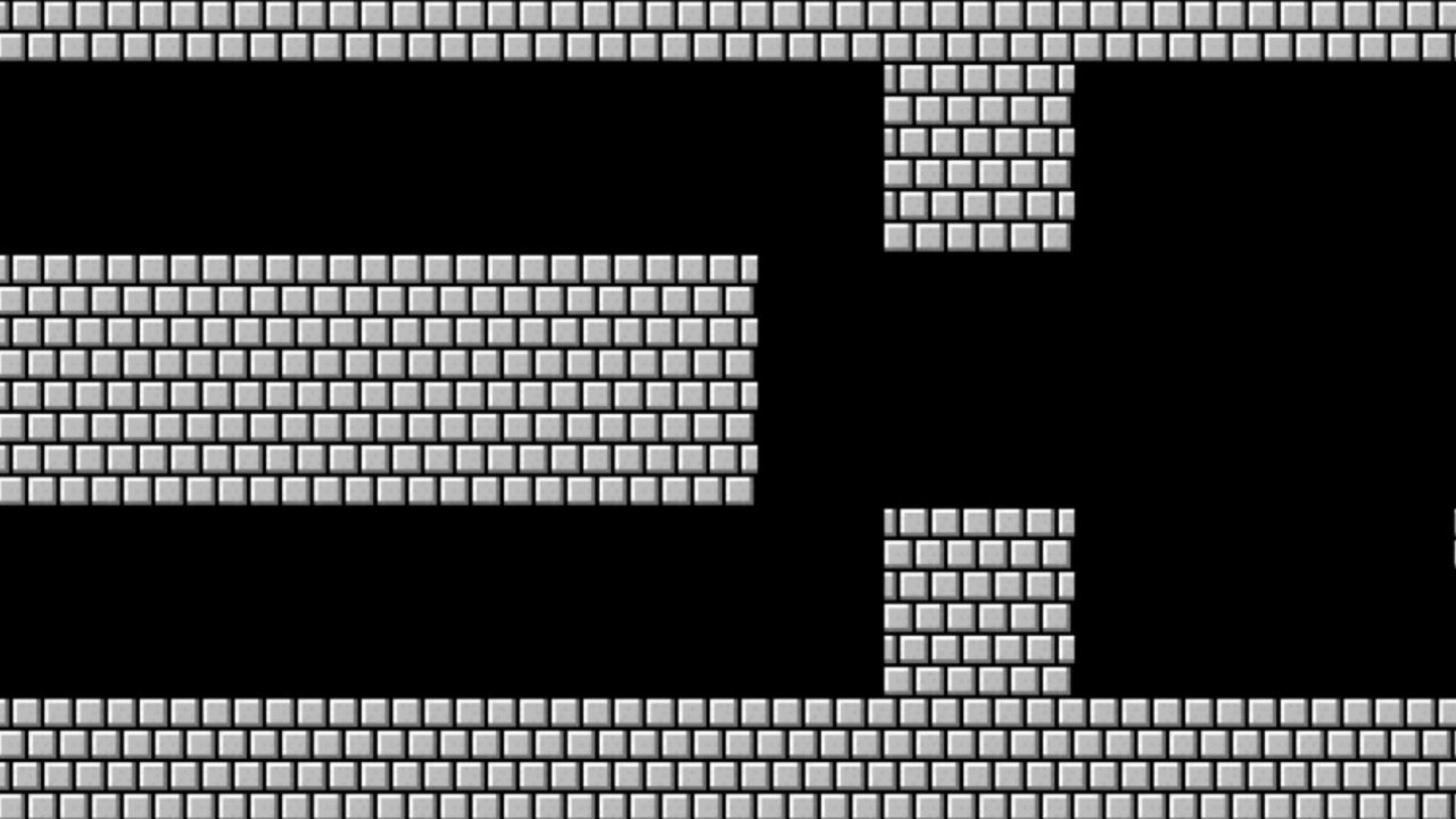


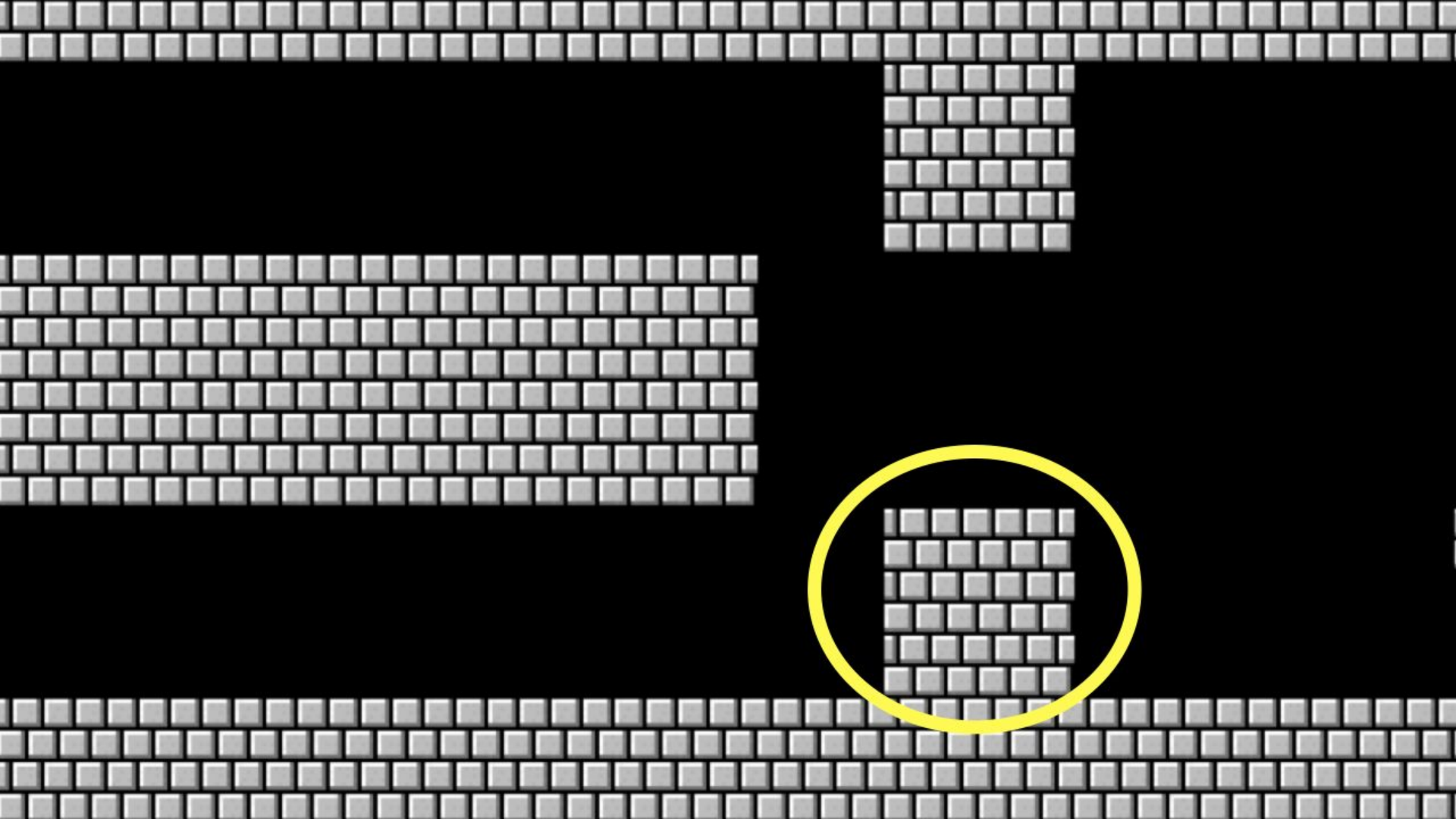






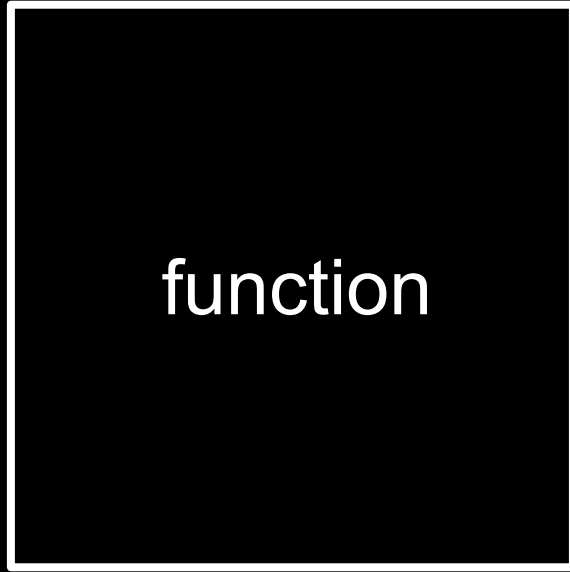






constants

arguments →



function

→ return value

+

-

\*

/

%

...

integer overflow



000



001

010

011

100

101

110

111

1000



000





truncation

floating-point imprecision

# Year 2000 Problem

1999

1999



1900

# Year 2038 Problem

19 January 2038

13 December 1901

19 January 2038



**This is CS50**